

EXAM 7 – SPRING 2014

11. (2.25 points)

For each of the following external systemic risks, give an example of a line of business for which the risk would be important, and explain why it is important for that line of business.

a. (0.75 point)

Legislative risk

b. (0.75 point)

Event risk

c. (0.75 point)

Latent claim risk

CONTINUED ON NEXT PAGE

SAMPLE ANSWERS AND EXAMINER'S REPORT

QUESTION 11	
TOTAL POINT VALUE: 2.25	LEARNING OBJECTIVE: A5: Describe the various sources of risk and uncertainty that are associated with the determination of reserves. Calculate risk margins that consider these sources of risk and uncertainty.
SAMPLE ANSWERS	
Part a: 0.75 points	
Sample LOB: • Workers Compensation • Liability • Auto • Medical Malpractice <i>Sample Answer 1</i> Legislative risk is important for Workers Compensation because law changes can affect benefit levels. If you don't account for this risk, you may not be able to handle increased exposure higher benefit levels. <i>Sample Answer 2</i> Workers Comp – there might be changes to benefit limits or items requiring coverage, which could effect in process or prior cases as well as future causing high variability in the reserves. <i>Sample Answer 3</i> Legislative risk is important for long-tail line such as Workers Comp. A change in benefit level could significantly impact the pricing/reserving of workers comp and such benefit change could be initiated by the state legislature. <i>Sample Answer 4</i> Legislative risk is very important for workers compensation. WC claims are very long tailed and can take a long time to pay out (possibly the life of a person.) Legislative changes to workers comp benefits may affect all open claims, even claims that occurred a long time ago, so reserves for many AYs could increase all at once.	

SAMPLE ANSWERS AND EXAMINER'S REPORT

Sample Answer 5

Medical Malpractice --> Court awards are often limited by caps set by law --> If there is tort reform, could change the cap, which would impact all open claims --> Thus the potential impact is huge. Also, the tort reform could have to do with the statute of limitations for a medical malpractice claims --> if the statute of limitations is extended, could mean more claims.

Part b: 0.75 points

Sample LOB:

- Property
- Homeowners
- Auto
- Earthquake

Sample Answer 1

Event risk (such as cats) is important for private passenger auto (property coverage) since event risk can impact many exposures at once. These risks may be highly correlated in tails between other lines and may be geographical correlated. If not diversified or properly capitalized, this can lead to insolvency.

Sample Answer 2

Property – important because an event like a catastrophe could have a significant impact on premium liabilities and on the solvency of the insurer. This type of risk can be modeled with external cat models for events like earthquake, hurricanes, etc.

Sample Answer 3

Homeowners – Event Risk. Homeowners writers can have many properties in a dense geographical area where one large weather event can affect all properties in the area. Thus, this is an important risk for the HO writers to consider.

Part c: 0.75 points

Sample LOB:

Several LOBs were accepted, the most common were:

- Workers Compensation
- Medical Malpractice
- Liability

Sample Answer 1

WC would be greatly impacted by latent claim risk. Since latent claims take a while to be discovered & WC is a long tailed line of business something like asbestos could be found to make ppl sick & the WC pol would have to pay even though didn't know this risk existed

SAMPLE ANSWERS AND EXAMINER'S REPORT

when policy was written.

Sample Answer 2

Latent – WC – think asbestos or WC occupational hazard type claims. Asbestos claims in particular over past 20 years have shown that have low freq but very very high severity claims have a large impact on estimating WC unpaid claim reserve. So important to capture latent risk for WC else would be missing key component risk.

Sample Answer 3

Latent claim risk is important for workers compensation. For asbestos for example, or other on the job work hazards, it can take years for health effects to be discovered. Insurers must consider the risk of latent claims for WC because the worker will be covered even if exposure to the substance happened a long time ago.

EXAMINER'S REPORT

- Candidates generally scored well with 50% receiving full or nearly full credit.
- Less than 10% had a point or more deducted.

Part a

- The candidate was expected to know **legislative** risk, what LOBs were sensitive, and why legislative risk is important.
- To obtain full credit, candidates were expected to respond with variations of:
 - Legislatures make law which determines what is compensable and statutorily-defined benefits, and thus impact claim/loss frequency, severity, loss cost.
 - Long-tail LOBs are more vulnerable to these changes, as the longer a claim's duration, the longer a claim's frequency, severity, and loss cost can be impacted by legislative change.
 - Legislatures *can* apply changes in statutory benefits that apply retroactively, and thus impact an array of open claims and AYs, which poses additional risk to the insurer.
- Common error made by candidates:
 - Confusing legislative risk with judicial, regulatory, or other government bodies. Legislatures make law and statutes; hence, statutory benefits. The judiciary interprets law and regulators implement law, but they are not the same as legislature.

Part b

- The candidate was expected to know **event** risk, what LOBs were sensitive, and why event risk is important.
- To obtain full credit, candidates were expected to respond with variations of:
 - A single event can trigger insurance coverage on a large number of policies all at once.
 - Property/Auto risks are typically independent (not correlated), but an event can trigger

SAMPLE ANSWERS AND EXAMINER'S REPORT

correlated claims and losses across a range of policies.

- Geographic concentration of risks can generate high frequency (when low frequency is the expectation).
- An event falls in the tail of distribution: low frequency and high severity.
- Consultation of risk models and subject matter experts is wise to consider exposure to event risk.

- Common error made by candidates:
 - Simply using “catastrophe” to define an event.
 - In and of itself, a large loss is not an event.

Part c

- The candidate was expected to know **latent claim** risk, what LOBs were sensitive, and why latent claim risk is important.
- To obtain full credit, candidates were expected to respond with variations of:
 - At policy inception, latent risks are unknown. These risks are not explicitly priced for and contract language is not explicitly aware of the exposure risk.
 - Latent risks take a long time to be discovered.
 - Latent risks are low frequency and typically emerge with high severity.
 - If a company is not adequately pricing and reserving for latent risks, then the company could be in adverse financial position.
- Common error made by candidates:
 - “Asbestos” is not a LOB.
 - Asbestos is an example of latent claim risk, but “asbestos” is not sufficient to explain why latent claim risk is important.
 - Just because a claim takes time to be reported does not mean that it is “latent”.